

Key Terminology

Load Balancer A component that distributes incoming traffic across multiple servers. Traditional tools such as NGINX and HAProxy rely on fixed routing rules like Round Robin and Least Connections.

Static Policies Rule-based load balancing strategies that do not change according to real-time system performance or workload fluctuations.

Telemetry Real-time system measurements such as CPU usage, memory consumption, queue length, response latency, and error rates. SmartLoad uses telemetry as an input for decision making.

Reinforcement Learning (RL) A learning-based approach where an agent improves routing decisions over time based on feedback such as latency and throughput.

Predictive Autoscaling Scaling decisions made in advance using forecasting models that predict future workload spikes rather than reacting only after high load occurs.

Forecasting Models Time-series techniques such as ARIMA, Prophet, or LSTM networks used to estimate upcoming traffic or resource demand.

Anomaly Detection Methods used to identify unusual or faulty system behavior, such as sudden delays or server instability. SmartLoad uses anomaly indicators to avoid routing traffic to unhealthy nodes.

Classical Algorithms Well-known routing approaches like Round Robin and Least Connections that act as a safe and interpretable baseline in SmartLoad.

Autoscaler A service (AWS AutoScaling, Azure VMSS, Google Autoscaler) responsible for adjusting the number of active servers based on load. These systems are typically reactive.

Orchestration Tools Platforms such as Kubernetes or Docker Swarm that manage containers, deployments, and scaling. They provide flexibility but add operational complexity.

Decision Layer The core logic of SmartLoad that combines RL, forecasting, telemetry, and anomaly detection to produce adaptive routing and scaling decisions.

Middleware A lightweight software layer—such as SmartLoad—that sits between incoming traffic and backend servers to provide intelligence, monitoring, and adaptive load management.